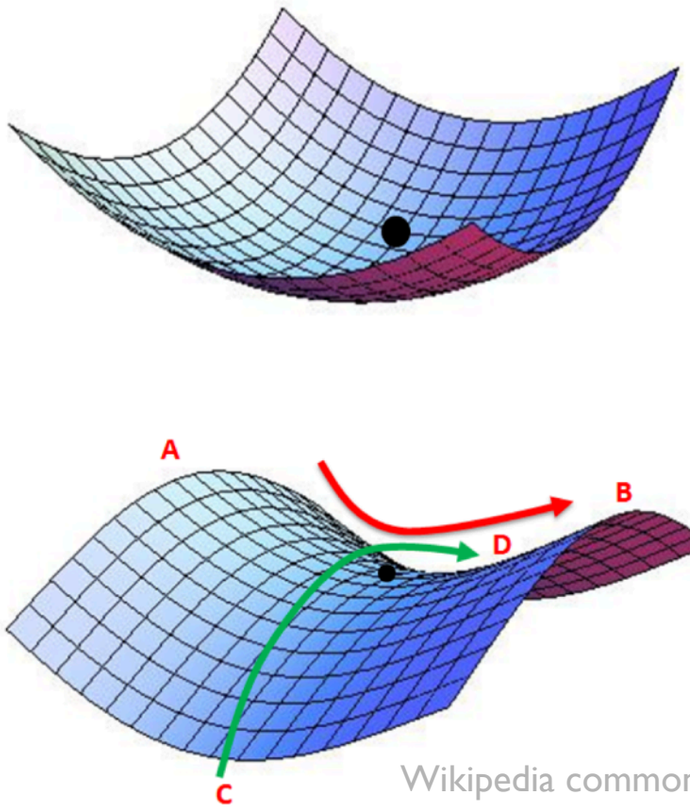
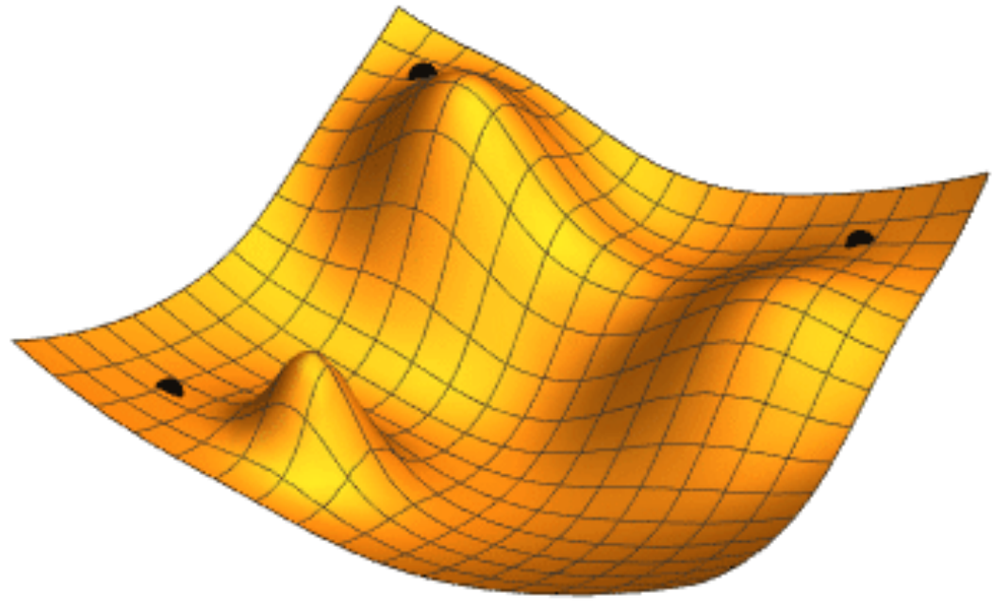
Week 7 - #1

Vector Analysis (I)



Wikipedia commons



gradient descent, Wikipedia commons

Today: Ch 6

Next Class: Ch 6

Ji-hoon Kim (Seoul National University)

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Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

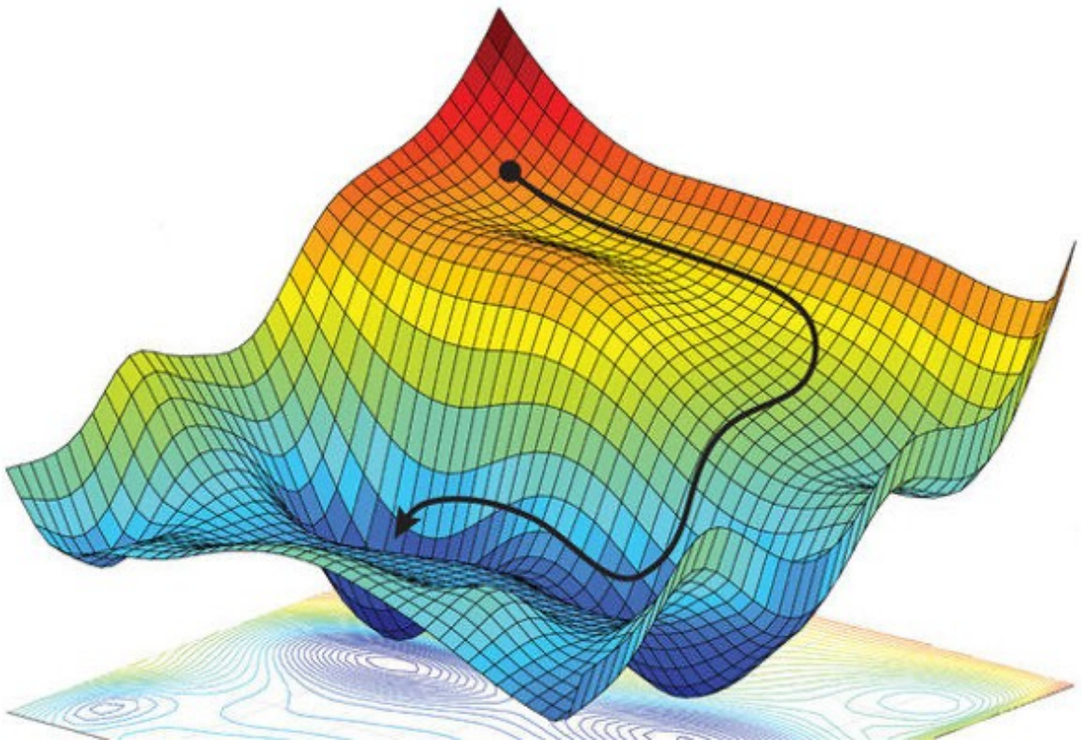
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #9

— [open book and open note, "and" cellphone or laptop, drop it off as you leave the class] —

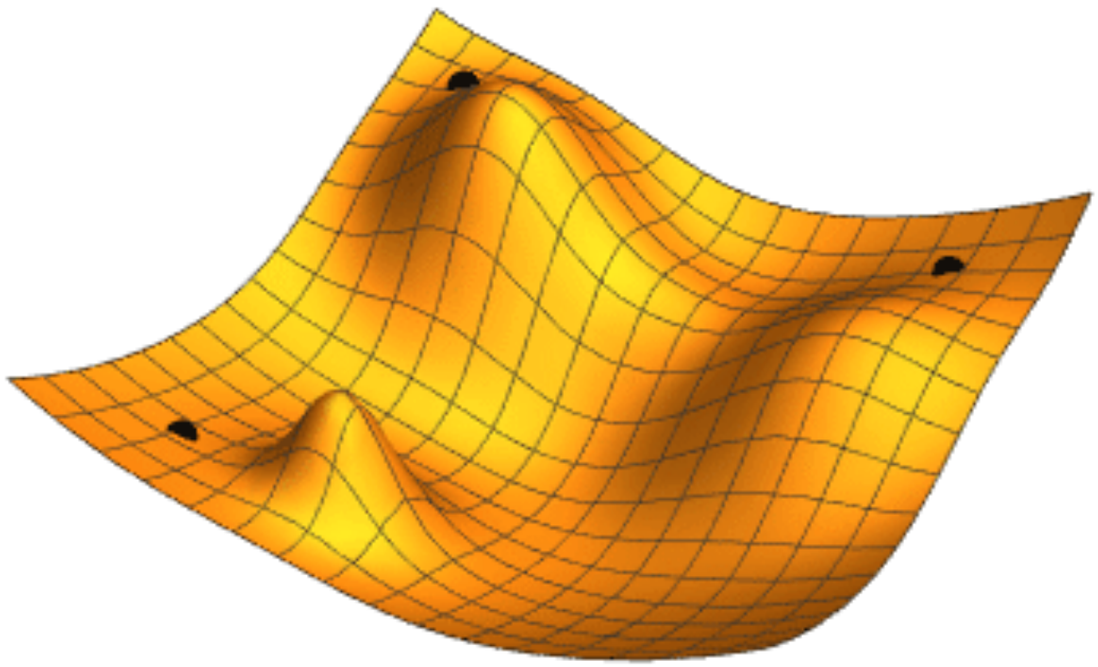
Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 5, Section 3, Problem 24

2. Explain in layman's terms what the figure/movie below are trying to visualize — by using the term "gradient". On a related note, study and explain briefly how the concept of "gradient" is used in machine learning and artificial intelligence.



Amini et al. (2018)



gradient descent, Wikipedia commons

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 6, Boas, 3rd ed.

The problems I suggest you to take a deeper look into include, but are not limited to, the following. The class homework assignments will mainly be from this list.

- Section 03: Problems 12, 13, 14, 16, 17
- Section 04: Problems 6, 9
- Section 06: Problems 8, 13, 14, 16
- Section 07: Problems 6, 7, 15, 16, 17
- Section 08: Problems 6, 14, 18, 19, 20
- Section 09: Problems 3, 4, 7, 8
- Section 10: Problems 6, 10, 12, 15, 16
- Section 11: Problems 8, 10, 12, 16, 17(b)(d)(e), 18
- Section 12: Problems 3, 16, 19

<u>Chapter 10, Sections 5 & 7-9</u>

- Section 05: Problems 5, 7, 10
- Section 08: Problems 6
- Section 09: Problems 2, 5

39 problems / 173 total

(Ch.10) 13 problems / 138 total